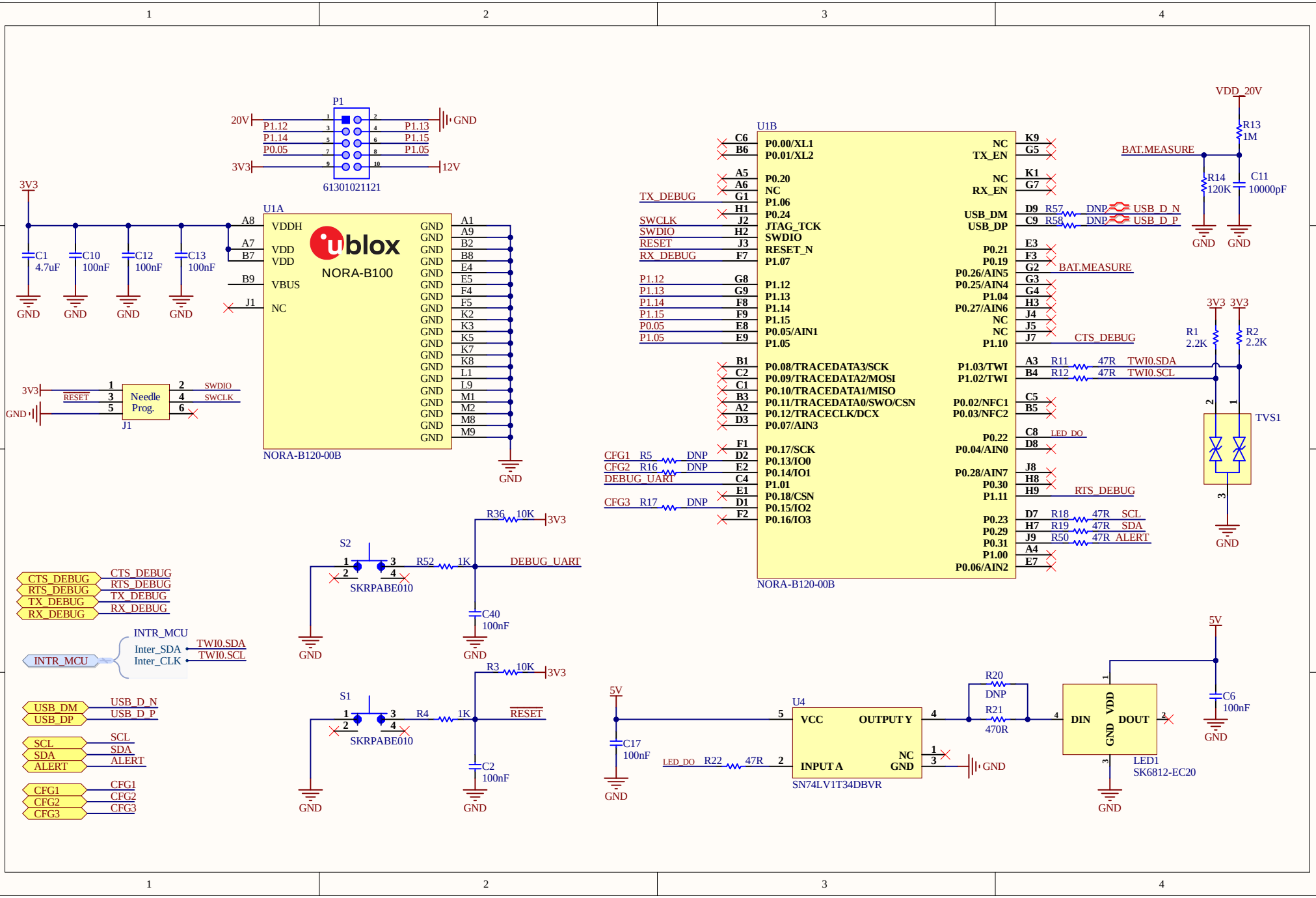
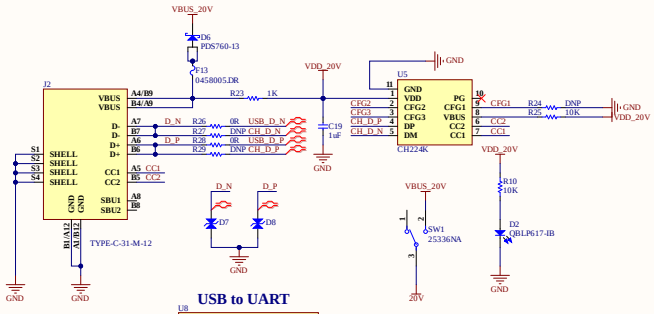
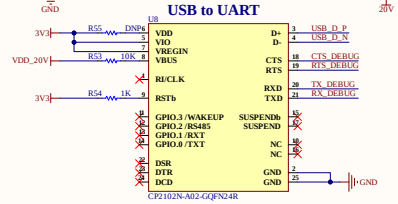
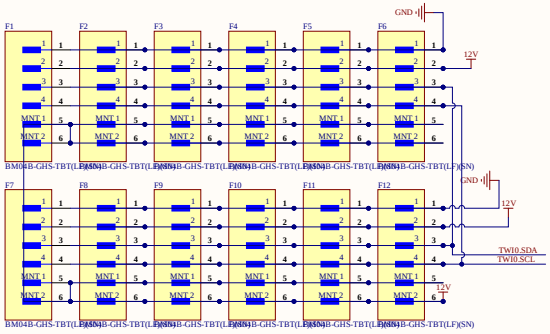

P1
61301021121
20V
3V3
GND
12V
P1.12
P1.13
P1.14
P1.15
P0.05
P1.05
U1A
NORA-B100
NORA-B120-00B
VDDH
VDD
VBUS
NC
C1 4.7uF
C10 100nF
C12 100nF
C13 100nF
J1
Needle Prog.
RESET
SWDIO
SWCLK
U1B
NORA-B120-00B
P0.00/XL1
P0.01/XL2
P0.20
NC
P1.06
P0.24
JTAG_TCK
SWDIO
RESET_N
P1.07
P1.12
P1.13
P1.14
P1.15
P0.05/AIN1
P1.05
P0.08/TRACEDATA3/SCK
P0.09/TRACEDATA2/MOSI
P0.10/TRACEDATA1/MISO
P0.11/TRACEDATA0/SWO/CSN
P0.12/TRACECLK/DCX
P0.07/AIN3
P0.17/SCK
P0.13/IO0
P0.14/IO1
P1.01
P0.18/CSN
P0.15/IO2
P0.16/IO3
TX_DEBUG
SWCLK
SWDIO
RESET
RX_DEBUG
DEBUG_UART
CFG1
CFG2
CFG3
R5 DNP
R16 DNP
R17 DNP
NC
TX_EN
RX_EN
USB_DM
USB_DP
P0.21
P0.19
P0.26/AIN5
P0.25/AIN4
P1.04
P0.27/AIN6
P1.10
P1.03/TWI
P1.02/TWI
P0.02/NFC1
P0.03/NFC2
P0.22
P0.04/AIN0
P0.28/AIN7
P0.30
P1.11
P0.23
P0.29
P0.31
P1.00
P0.06/AIN2
BAT.MEASURE
CTS_DEBUG
LED_DO
RTS_DEBUG
TWI0.SDA
TWI0.SCL
SCL
SDA
ALERT
R57 DNP
R58 DNP
USB_D_N
USB_D_P
R11 47R
R12 47R
R18 47R
R19 47R
R50 47R
VDD_20V
R13 1M
R14 120K
C11 10000pF
R1 2.2K
R2 2.2K
TVS1
S2
S1
SKRPABE010
R36 10K
R52 1K
R3 10K
R4 1K
C40 100nF
C2 100nF
INTR_MCU
Inter_SDA
Inter_CLK
CFG1
CFG2
CFG3
U4
SN74LV1T34DBVR
VCC
OUTPUT Y
INPUT A
NC
R20 DNP
R21 470R
R22 47R
C17 100nF
C6 100nF
5V
DIN
VDD
GND
DOUT
LED1
SK6812-EC20

USB AND POWER

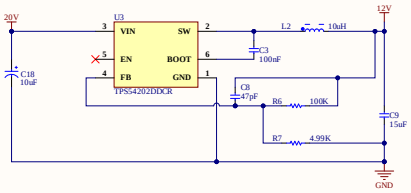


USB to UART

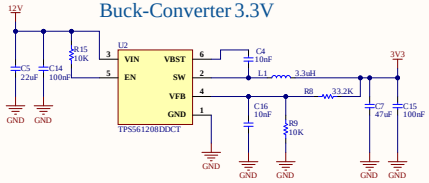




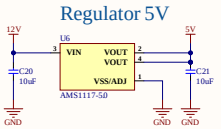
Buck-Converter 12V



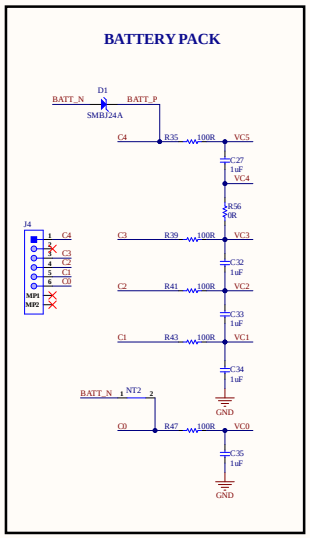
Buck-Converter 3.3V

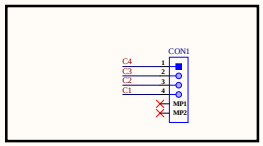


Regulator 5V

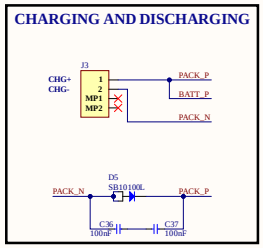


BATTERY PACK



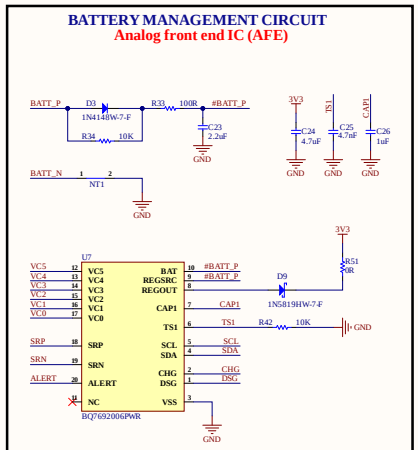


CHARGING AND DISCHARGING

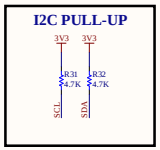


BATTERY MANAGEMENT CIRCUIT

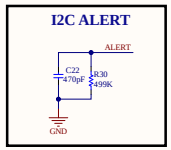
Analog front end IC (AFE)



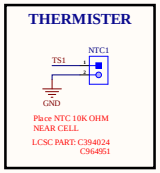
I2C PULL-UP



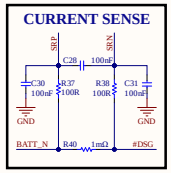
I2C ALERT



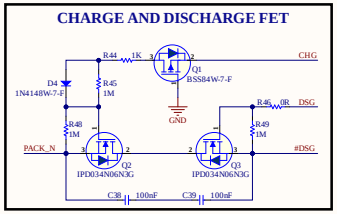
THERMISTER

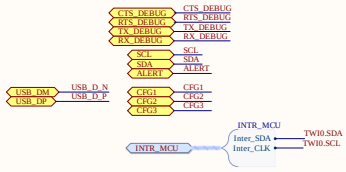


CURRENT SENSE



CHARGE AND DISCHARGE FET





Title		
Size: A2	Number	Revision
Date: 11/25/2025	Sheet of	
File: D:\eSpark-Consultants\...\Piezo_Regulator_Board\PowerSch.SchDoc		

